

Energy Balancing for Revenue Protection - TGI Use Case Notes



[UC28] - Awesense Platform Use Case

Overview

This manual is intended to:

- Provide instructions to create the selected use case in True Grid Intelligence (TGI).

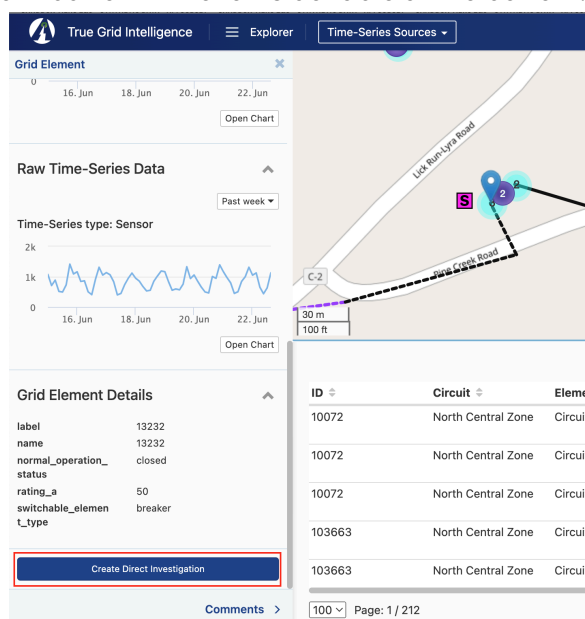
For more details about Energy Balancing for Revenue Protection and how it can be analyzed using Awesense Platform, please refer to the [UC28 - Energy Balancing for Revenue Protection.pdf](#) document.

TGI Use Case Notes

This document outlines the implementation of the mentioned use case in the TGI visualization interface, along with instructions for creating a similar use case on your data.

1. New energy balance:

The usual way of getting the energy balance in TGI is to create an investigation. This can be made, for example, in Explorer (select from the main menu - top hamburger icon)¹. A measurement sensor with measured data must be found, and a Placement must be created at the location of this sensor. Select the measuring sensor from the map or table, then click the "Create Direct Investigation" button in the left side table at the bottom.



¹ There are other ways of creating investigation in TGI like for example:

- Placement Page and then creating direct investigation;
- Segment Analyzer Page and then creating investigation from segment;
- Investigation Page and then creating sub-investigation.

Energy Balancing for Revenue Protection - TGI Use Case Notes

[UC28] - Awesense Platform Use Case



In the Investigation, the start date has to be set. To calculate the energy balance, it is necessary to select the segment for which we want to calculate the balance. In this case, it is the Target Segment. So in the left sidebar, we click on the "Target Segment" button and then on the "Recalculate Energy Balance" button.

The screenshot displays the True Grid Intelligence (TGI) Awesense platform interface. The top navigation bar includes the TGI logo, the text "True Grid Intelligence", a menu icon, the path "Investigations » Investigation of 39660", and a "Map" dropdown button.

The left sidebar contains the following sections:

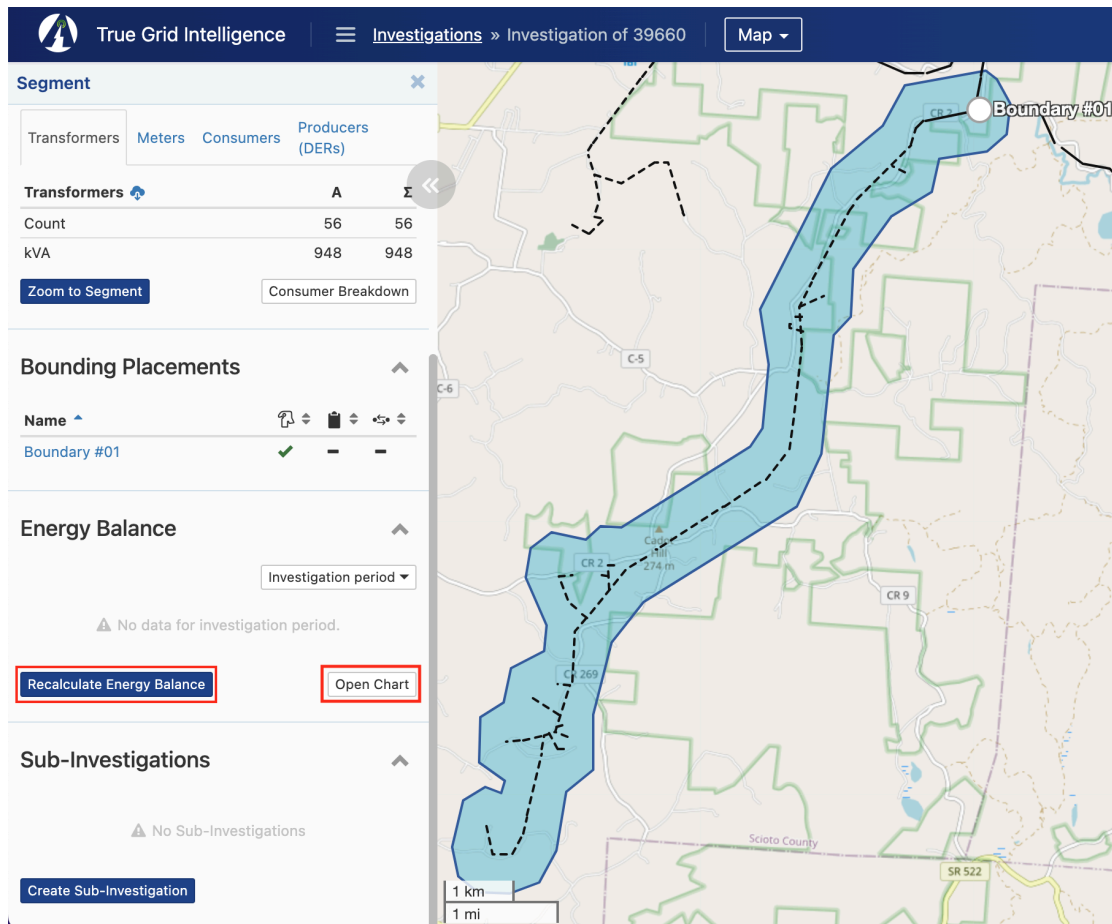
- Circuit:** North Central Zone
- GIS Date:** 1 Jan, 2021 9:00 PM
- Owner:** [awesense-petr](#)
- Starts:** 1 Jun, 2023 5:23 PM (highlighted with a red box)
- Description:** No Description
[Show more / Update](#)
- Target Area:**
 - Transformers: 56
 - Meters: 948
 - Consumers: 948
 - Producers (DERs): 0

Buttons: [Zoom to Investigation](#), [Consumer Breakdown](#)
- Segments:**
 - [Target Segment](#) (highlighted with a red box)

The main map area shows a geographical view of the investigation area. A blue-shaded region represents the target area. A dashed line indicates the boundary of the target area, labeled "Boundary #01". The map includes labels for various roads (C-5, C-6, CR 2, CR 9, SR 522) and a scale bar (1 km, 1 mi). The map also shows a "Scioto County" label.

Energy Balancing for Revenue Protection - TGI Use Case Notes

[UC28] - Awesense Platform Use Case



Once calculated, the Energy Balance is displayed. To explore this balance in more detail, click on the "Open Chart" button.

Energy Balancing for Revenue Protection - TGI Use Case Notes

[UC28] - Awesense Platform Use Case



Energy Balance

Discrepancy: 61,293 kWh (46.7%), 3,284 kWh/day (18.7 days)

Recalculated on 22 Jun, 2023 5:34 PM

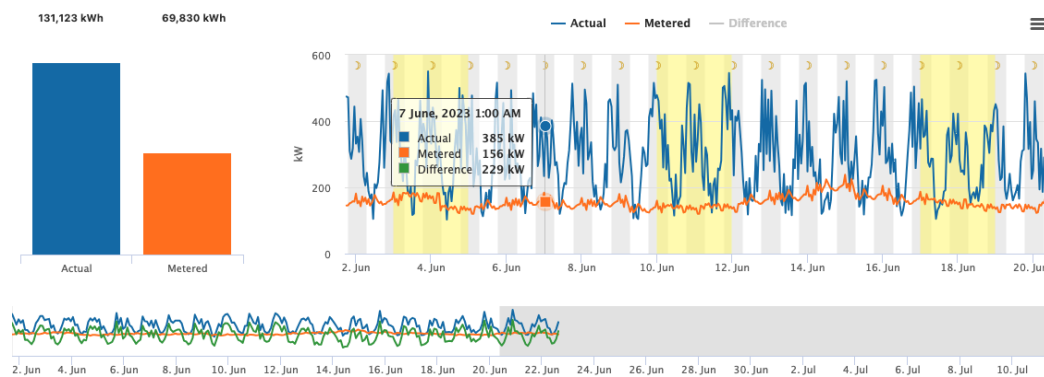
Recalculate Energy Balance

1 Jun, 2023 - 20 Jun, 2023

Show More Options

ACTUAL VS METERED

PROFILE



The energy balance graph shows the energy difference between the measured points and the real consumption. The basic interface of the energy balance chart allows you to change the different time periods of the energy balance. To quickly assess the state, summary information is provided on the total discrepancies and daily losses.

Using the "Show More Options" button, it is possible to display other supporting indicators, such as Show Temperature, Show Weekends, Show Nights, etc., for easier analysis of the values.

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[UC28] - Awesense Platform Use Case



Energy Balance

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Recalculated on 22 Jun, 2023 5:34 PM

Recalculate Energy Balance

1 Jun, 2023 - 20 Jun, 2023

Show Less Options

Units
kWh

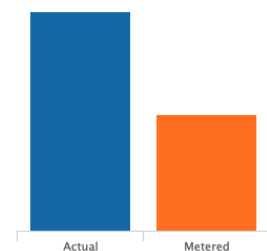
Chart Type
Line Chart

☒ Show Nights
☒ Show Weekends
☐ Show Temperature

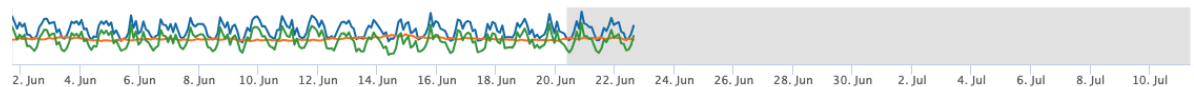
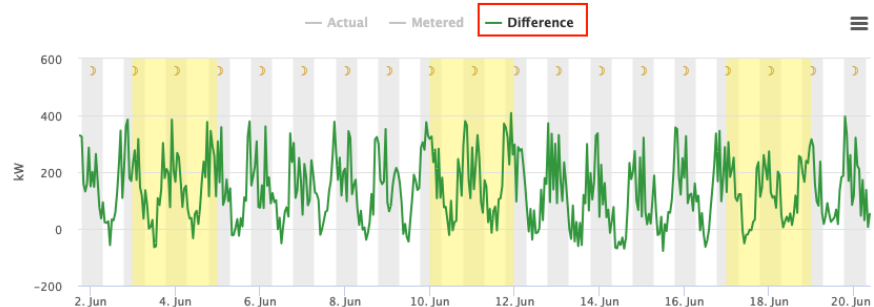
ACTUAL VS METERED

131,123 kWh

69,830 kWh



PROFILE



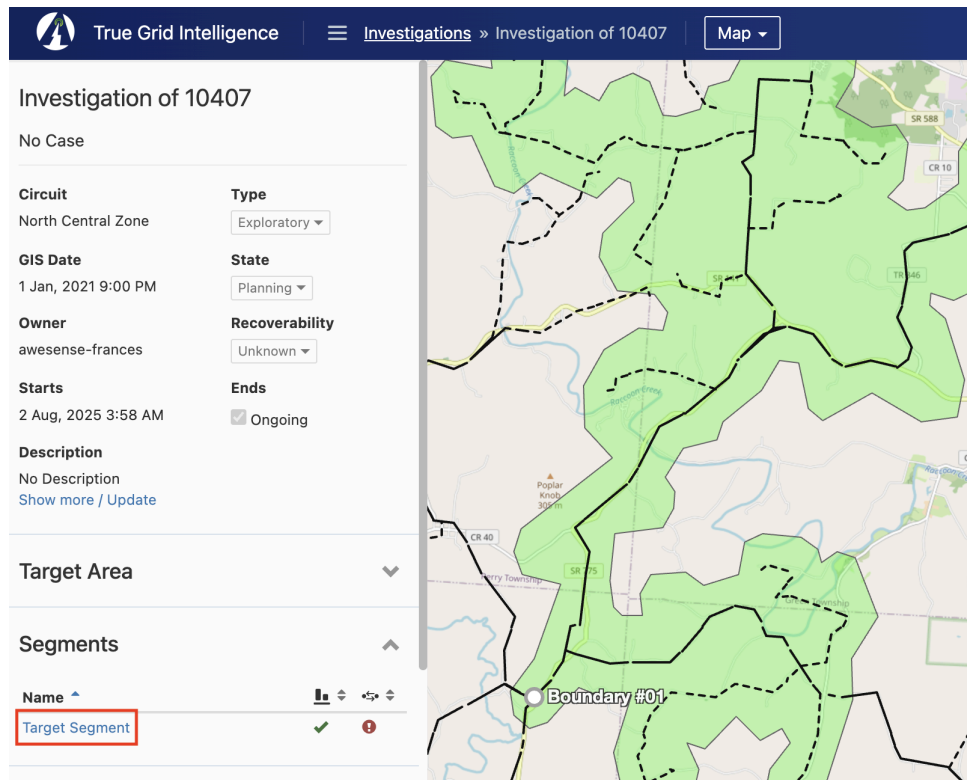
Using the "Difference" button, you can display the difference in these measured energies over time and easily analyze when unauthorized consumption occurs. Using this feature, some signature patterns can be identified, and the source of losses can be revealed more easily.

2. Existing energy balance:

Existing energy balances can be found in the Investigations menu (select from the main menu - top hamburger icon). Select one of the created investigations. It is necessary to choose the segment to which you want to see the balance. In this case, it is the target segment. In the left sidebar, click on the "Target Segment" button.

Energy Balancing for Revenue Protection - TGI Use Case Notes

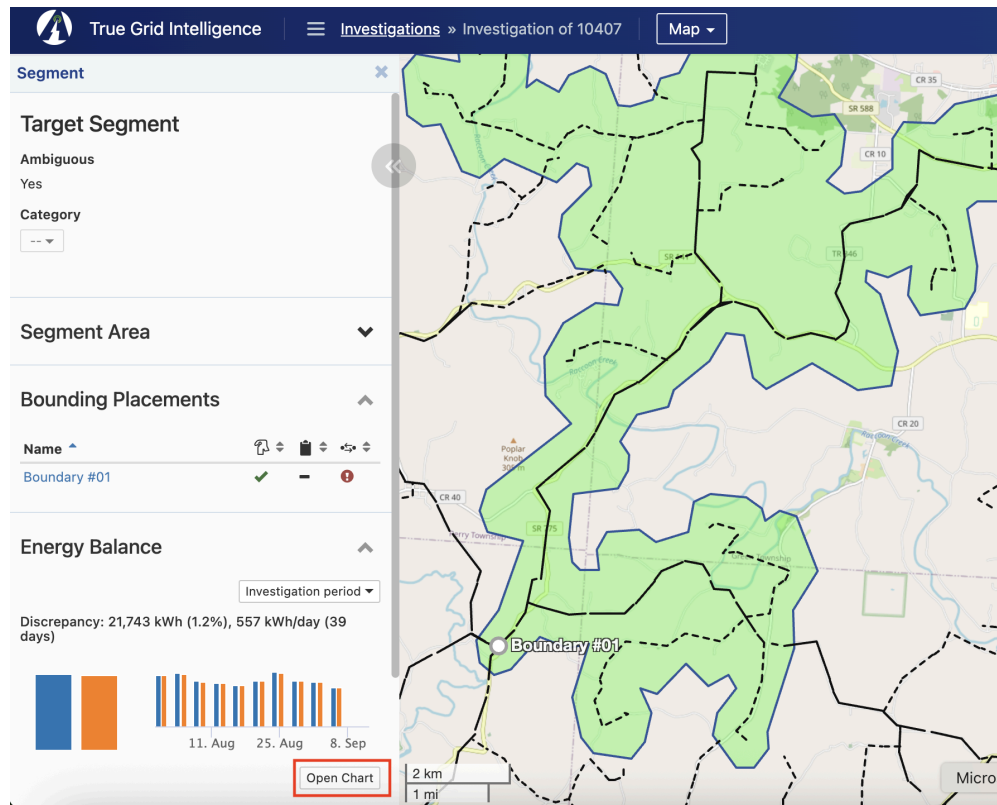
[UC28] - Awesense Platform Use Case



The energy balance is displayed at the bottom left. To explore this balance in more detail, click on the "Open Chart" button.

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[UC28] - Awesense Platform Use Case



The next steps are the same as described earlier in the document in the section 1.

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[UC28] - Awesense Platform Use Case

General Note

- This description was created for TGI version 3.16.0. For future versions of TGI, the process might be slightly different.
- Access to the different parts of TGI varies depending on the user's permissions.
- Users on the Sandbox environments currently have read-only permissions; therefore, this use case may be accessible only for viewing, not creation or editing. If you are interested in the latter, please contact Awesense.
- Some of the data for this use case may be available only on Sandbox tiers 2 & 3, but not on tier 1. If you are interested in upgrading your tier access, please contact Awesense.
- For more information, please contact tools@awesense.com.